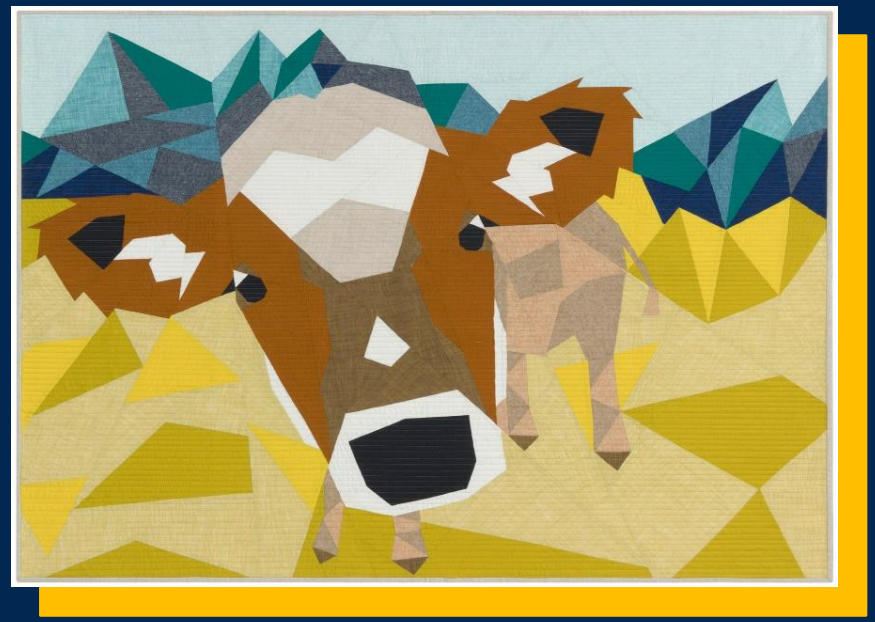
MPM 200 · WEEK 3

Introduction to R Programming and Collaborative Science





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Learning objectives



Introduction to R programming

Basics of R syntax and data types



Project management

Setting up Google Colab and your coding environment



Collaborative and reproducible science

Best practices for collaborative research and reproducibility

Today's game plan

1

Lecture

Why data management and reproducibility matter, and how to do them

2

R Quest (live demo)

Introduction to R.ipynb in Colab: 6 levels and a boss level, ending with an epi curve

3

Assignment

Will the Hummers Survive? Rescue records from California wildlife rehab centers

Before the demo: open the notebook from the course GitHub with the Open in Colab badge and switch the runtime to R.

Data is valuable

Great effort goes into collecting data systematically.

Hard work, meticulous planning and resources go into every dataset: field teams, permits, lab time, funding.

Losing, corrupting or being unable to re-use data wastes all of that effort.



Why data management?

Safe storage and sharing

Data that survive a broken laptop, a graduate, and a reviewer's request.

Reproducible research

Anyone, including future you, can re-run the analysis and get the same answer.



Alston JM & Rick JA (2021). A beginner's guide to conducting reproducible research. Bull Ecol Soc Am 102(2):e01801.

Five habits of good data management



Organize

your project folder



Protect

your raw data



Name

your files effectively



Track

your project's changes



Back up

your project

The next slides take these one at a time.

Anatomy of a working directory

1. **Start every project in its own folder**

   In RStudio, an .Rproj file; in Colab, a project folder you create.

2. **Keep raw data immutable**

   Never overwrite it. Clean copies go in a separate folder.

3. **Modularize code by task**

   One notebook or script per job: cleaning, analysis, figures.

4. **Store outputs separately**

   Plots and summaries go in outputs/, never next to raw data.

README MIT license

```
├── LICENSE
├── Makefile           <- Makefile with commands like `make setup` or `make conda-create`
├── README.md          <- The top-level README for developers using this project.
├── data
│   ├── external       <- Data from third-party sources.
│   ├── interim        <- Intermediate data that has been transformed.
│   ├── processed      <- The final, canonical data sets for modeling.
│   └── raw            <- The original, immutable data dump.
├── models             <- Trained and serialized models, model predictions, or model summaries
│
├── notebooks          <- Jupyter notebooks. The naming convention is a date (for ordering),
│                         the creator's initials, and a short `-` delimited description, e.g.
│                         `03132024-pranav-data-exploration`.
│
├── references         <- Data dictionaries, manuals, and all other explanatory materials.
│
├── reports            <- Generated analysis as HTML, PDF, LaTeX, etc.
│   └── figures        <- Generated graphics and figures to be used in reporting
│
├── environment.yml   <- The environment file for reproducing the analysis environment, e.g.
│                         generated with `conda create -f environment.yml`
│
├── src                <- Source code for use in this project.
│   ├── __init__.py    <- Makes src a Python module
│   │
│   ├── data           <- Scripts to process data
│   │
│   ├── models         <- Scripts to train models and then use trained models to make
│   │   │                 predictions
│   │
│   └── visualization  <- Scripts to create exploratory and results-oriented visualizations
```

Naming files

No spaces

Use snake_case or kebab-case

Order with prefixes

01_, 02_, 03_ for sequential steps

Date or version it

YYYYMMDD sorts correctly; or v1.0

Examples

```
20251020_west_nile_cleaning.R
```
✓

```
bird_migration_2024_raw.csv
```
✓

```
Final data (2) NEW.xlsx
```
✗

Why bother?
Files sort themselves in order, computers can read them, and you can tell what each one is in 6 months.

Naming objects in R

Pick one style and stick to it

camelCase `totalCases`

snake_case `total_cases`

Embed meaning in the name

`_df` data frames, e.g. covid_df

`_mat` matrices, e.g. contact_mat

Descriptive

Consistent

Human readable

Machine readable

Data management and backup

Raw ≠ derived

Store raw and derived data in separate folders.

Keep a data dictionary

e.g. data_description.xlsx: what each column means and its units.

Record metadata

Where the data came from and every processing step.

Back it up

Box, Google Drive, GitHub, or rsync/Backblaze. Ideally more than one.

Version control your data and code



Git + GitHub

Every change is recorded and reversible. github.com



Cloud drives

Google Drive, Box or Dropbox: sync plus file version history.



External hard disk

A local copy that doesn't need the internet.

In this course: save notebooks to your own branch of the EpiPandit/MPM_200 repository. Never to main.

A reproducibility checklist

1 Before analysis

Raw data backed up in several places, in open formats, with metadata.

2 During analysis

Code clean, commented, reviewed by a colleague; software versions documented.

3 After analysis

Clear instructions to find data and code; shared together at a permanent site.

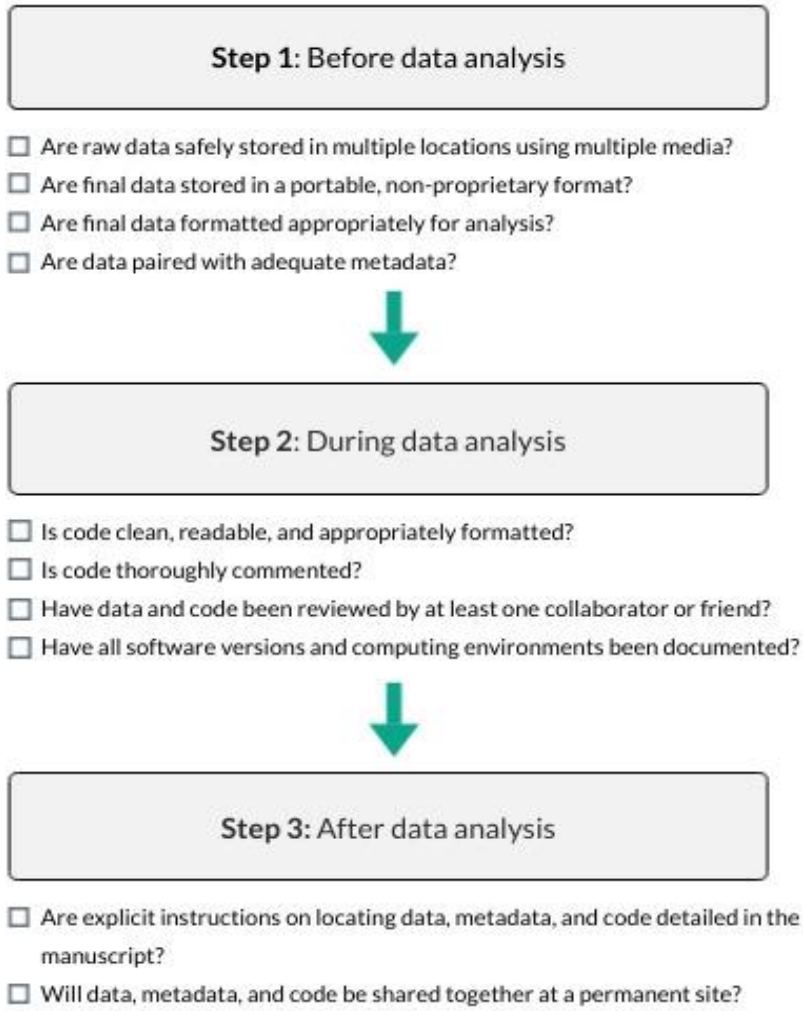


Figure: Alston JM & Rick JA (2021). Bull Ecol Soc Am 102(2):e01801. doi:10.1002/bes2.1801

Your coding environment: Google Colab

R in the browser. Nothing to install, easy to share.



Get R running in Colab in five steps

1

Sign in

Go to Google Colab with your UC Davis Google account

2

Open the notebook

Click Open in Colab on the course GitHub, or File → New notebook in Drive

3

Switch to R

Runtime → Change runtime type → R. Leave hardware on CPU.

4

Load packages

install.packages() and library() at the start of every session

5

Save your work

File → Save a copy in GitHub, to your own branch. Date-stamp the file name.

Full walkthrough: Google Colab Setup Instructions.pdf on the course GitHub

Colab gotchas (save yourself an hour)



Colab forgets

Packages, uploaded files and objects vanish when the runtime disconnects. Re-run from the top.



You live in /content

Paths from someone else's laptop (setwd("C:/Users/...")) will always fail. Use relative paths.



beep() is silent

beep() runs on Google's machine. The assignment includes a helper that plays sound in your browser.



Never save to main

Save a copy in GitHub to your own branch. Main belongs to the instructors.

Live demo: R Quest

Introduction to R.ipynb · You're a new epidemiologist in August 2020 with five months of state COVID-19 reports to explain by Friday.

- **0** Level 0 · Set up Colab
- **1** Level 1 · R as a calculator: objects, data types, vectors
- **2** Level 2 · Packages (and a talking cow)
- **3** Level 3 · Working directory and a tidy project
- **4** Level 4 · Meet the data, plus a bug hunt
- **5** Level 5 · Wrangle with dplyr
- **6** Level 6 · Build a ggplot layer by layer
- ★ **Boss level · California's epi curve**

Your mission: Will the Hummers Survive?

6,908 hummingbird rescues from three California rehab centers, 1991–2016

Five case files, a data-detective bonus, and a plot hall of fame

